4. Entire functions

Jensen formula (D_R , C_R be the disc and circle centered at zero with radius R)

Thm I.4.1. Let Ω be an open set containing $\overline{D_R}$. Suppose $f \in H(\Omega)$, $f(0) \neq 0$

$f(z) \neq 0$ $\forall z \in C_R$. If $z_1, \dots z_N$ denote the zeros of f inside the disc, (the finiteness is guaranteed by Argument Principle)

then $$\log|f(0)| = \sum_{k=1}^{N} \log\left(\frac{|z_k|}{R}\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta$$

proof. First we say that if f_1, f_2 satisfies the Prop, so is $f_1 f_2$

Second, write $f(z) = (z-z_1)\dots(z-z_N)\, g(z)$ $g(z) \in H(\Omega)$. and never vanishes on $\overline{D_R}$.

So we just need to prove ~~that~~ cases of g and $z-w$

① Since $g(z) \neq 0$ $\forall z \in \overline{D_R}$. we can write $g(z) = e^{h(z)}$.

Thus $|g(z)| = e^{\mathrm{Re}(h(z))} \Rightarrow \log|g(z)| = \mathrm{Re}(h(z))$

By mean-value theorem $h(0) = \frac{1}{2\pi}\int_0^{2\pi} h(Re^{i\theta})\, d\theta \Rightarrow \mathrm{Re}(h(0)) = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{Re}(h(Re^{i\theta}))\, d\theta$

② $f(z) = z-w$ $(w \in D_R)$ We have $\int_0^{2\pi} \log\left|e^{i\theta} - \frac{w}{R}\right| d\theta = \int_0^{2\pi} \log\left|1 - \frac{w}{R}e^{i\theta}\right| d\theta$

let $F(z) = 1 - \frac{w}{R}z$ $(z \in D_R) \Rightarrow F(z) \neq 0$ $\forall z \in D_R$

$\Rightarrow \exists\, G(z)$ s.t. $F(z) = e^{G(z)}$. $\Rightarrow \log|F(z)| = \mathrm{Re}(G(z))$

Since $F(0) = 1$, we have $0 = \frac{1}{2\pi}\int_0^{2\pi} \log\left|1 - \frac{w}{R}e^{i\theta}\right| d\theta$ □

Def I.4.2. $n(r)$ be the number of zeros inside the disc D_r.

Lemma I.4.3. If $z_1, \dots, z_N$ are zeros inside the disc D_R, $f(0) \neq 0$, $f \in H(\Omega)$. then $\int_0^R n(r)\frac{dr}{r} = \sum_{k=1}^{N} \log\left|\frac{R}{z_k}\right|$.

proof: $\sum_{k=1}^{N} \log\left|\frac{R}{z_k}\right| = \sum_{k=1}^{N} \int_{|z_k|}^{R} \frac{dr}{r} \xlongequal[\eta_k(r)=\begin{cases}1 & r>|z_k| \\ 0 & r\leq|z_k|\end{cases}]{} \sum_{k=1}^{N}\int_0^R \eta_k(r)\frac{dr}{r} = \int_0^R \left(\sum_{k=1}^{N}\eta_k(r)\right)\frac{dr}{r} = \int_0^R n(r)\frac{dr}{r}$ □

Coro I.4.4. $$\int_0^R n(r)\frac{dr}{r} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta - \log|f(0)|$$ □

Now we can estimate functions of finite order.

Def I.4.5. f be entire, if $\exists\ \rho > 0$ & $A, B > 0$. such that $|f(z)| \leq A e^{B|z|^{\rho}}$ $\forall z \in \mathbb{C}$.

then we say that f has an order of growth $\leq \rho$.

The order of f : $\rho_f = \inf \rho$

Thm I.4.6. Let f be entire with order $\leq \rho$, then: (i) $n(r) \leq C r^{\rho}$. for some $C > 0$ and large r.

(ii) If $z_1, z_2, \dots$ denote zeros of f, $z_k \neq 0$. then $\forall s > \rho$. we have $\sum_{k\geq 1} \frac{1}{|z_k|^s} < \infty$.

proof: (i) W.L.O.G $f(0) \neq 0$ (otherwise we consider $\frac{f(z)}{z^{l}}$. l be the order of zero at the origin).

Then by Coro I.4.4. $\int_0^R n(x)\frac{dx}{x} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta - \log|f(0)|$

$\Rightarrow \int_r^{2r} n(x)\frac{dx}{x} \leq \frac{1}{2\pi}\int_0^{2\pi} \log|f(R e^{i\theta})|\, d\theta - \log|f(0)|$

Besides, $\int_r^{2r} n(x)\frac{dx}{x} \geq n(r)\int_r^{2r}\frac{dx}{x} = n(r)\log 2$

$$\int_0^{2\pi} \log|f(Re^{i\theta})|\,d\theta \le \int_0^{2\pi} \log|Ae^{BR^{\rho}}|\,d\theta \le C'r^{\rho} \qquad (R=2r) \text{ for all large } r$$

Thus $n(r) \le C r^{\rho}$

(ii) ~~$\sum_{|z_k|\le 1}$~~ The nambor of zeros in the disc is finite (otherwise $f \equiv 0$)

So it saffices to chow that $\sum_{|z_n|\le 1} |z_k|^{-s}$ finite

$$\sum_{|z_n|\ge 1} |z_k|^{-s} = \sum_{j=0}^{\infty}\Big(\sum_{2^j\le|z_k|\le 2^{j+1}} |z_k|^{-s}\Big) \le \sum_{j=0}^{\infty} 2^{-js}\, n(2^{j+1})$$

$$\le c\sum_{j=0}^{\infty} 2^{-js}\, 2^{(j+1)\rho} \le c' \sum_{j=0}^{\infty} 2^{(\rho-s)j} < \infty \quad (\rho<s)$$

□

Prop I.4.7. $\{F_n\} \subseteq H(\Omega)$. If $\exists\, c_n>0$. such that $\sum_{n\ge1} c_n<\infty$. $|F_n(z)-1| \le c_n \quad \forall z\in\Omega$

Then (i) $\prod_{n\ge1} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$

(ii) If $F_n(z) \ne 0$. $\forall n\in\mathbb{N}_+$. then $\dfrac{F'(z)}{F(z)} = \sum_{n\ge1}\dfrac{F_n'(z)}{F_n(z)}$.

proof: (i) $F_n(z) = 1+a_n(z)$. Then $\sum_{n\ge1}|a_n(z)| \le \sum_{n\ge1} c_n < \infty$ Thus $\sum a_n(z)$ converges uniformly.

Hence $\prod_{n\ge1} F_n(z)$ converges uniformly. to a holomorphic function $F(z)$. (Here we use Morera's Thm

(ii) . $\forall\, K\subseteq\Omega$ compact, we have $\underset{G_n(z)}{\underset{\parallel}{\prod_{1\le n\le N}F_n(z)}} \to F(z)$ converges uniformly on K.

Then $G_n'(z) \to F'(z)$ converges uniformly on K $\Rightarrow \dfrac{G_n'}{G_n} \to \dfrac{F'(z)}{F(z)}$ ($G_n(z)\ne 0$ for n large)

Here $\dfrac{G_n'(z)}{G_n(z)} = \sum_{n\ge k\ge1} \dfrac{F_k'(z)}{F_k(z)} \Rightarrow \dfrac{F'(z)}{F(z)} = \sum_{n\ge1}\dfrac{F_n'(z)}{F_n(z)}$ □

Then we'll show the famous Weierstrass canonical factorization theorem

Thm I.4.8. Given $\{a_n\} \subseteq \mathbb{C}$ with $|a_n| \to \infty$. Then $\exists \underset{!!!}{\underline{f \text{ entire}}}$ only vanishes at all $z = a_n$. ① And other such entire function is of the form $f(z)e^{g(z)}$ where g is entire ②

proof: ② is obvious. Since $\frac{f_1}{f_2}$ is entire and $\frac{f_1(z)}{f_2(z)} \neq 0 \ \forall z \in \mathbb{C}$

Then $\exists g(z) \in H(\mathbb{C})$ s.t. $\frac{f_1(z)}{f_2(z)} = e^{g(z)}$

①: Now we define canonical factors by $E_0(z) = 1 - z$, $E_k(z) = (1-z)\, e^{z + \frac{z^2}{2} + \dots + \frac{z^k}{k}}$ for $k \geq 1$

Lemma I.4.9. If $|z| < \frac{1}{2}$, then $|1 - E_k(z)| \leq c|z|^{k+1}$ for some $c > 0$

proof: $E_k(z) = e^{\log(1-z) + z + \frac{z^2}{2} + \dots + \frac{z^k}{k}} = e^{w}$.

Where $w = -\sum_{n \geq k+1} \frac{z^n}{n} \Rightarrow |w| \leq |z|^{k+1} \sum_{n \geq k+1} \frac{|z|^{n-k-1}}{n} < 2|z|^{k+1}$

Thus $|1 - E_k(z)| = |1 - e^{w}| \leq e|w| \leq 2e|z|^{k+1}$ $\quad$ ($|1 - e^{w}| \leq e|w|$ is from $|1 - e^{w}| \leq \sum_{k \geq 1} \frac{|w|^k}{k!}$ $= |w| \sum_{k \geq 1} \frac{|w}{k}$ $< e|w|$)

Back to the proof: We define $f(z) = z^m \prod_{n \geq 1} E_n\left(\frac{z}{a_n}\right)$

Fix $R > 0$. $\forall |z| < R$, we prove the properties that f ought to have.

There're finitely ~~∅~~ a_n s.t. $|a_n| \leq 2R$. the responding finite product vanishes at all $z = a_n$. with $|a_n| < R$

If $|a_n| \geq 2R$, then $\left|\frac{z}{a_n}\right| < \frac{1}{2}$ by I.4.9. We have $\left|1 - E_n\left(\frac{z}{a_n}\right)\right| < \frac{c}{2^{n+1}}$. (Actually $c = 2e$)

Therefore $\prod_{|a_n| \geq 2R} E_n\left(\frac{z}{a_n}\right)$ is holomorphic on D_R and doesn't vanish on D_R.

Hence we ~~pr~~ construct the desired ~~th~~ function $\square$.